

Modular Top Down RTS Camera

Introduction:

This is a modular RTS camera controller. Each camera control feature is a component. Each component can be turned on by simply applying on camera and turned off by removing the component.

Configurations are subdivided into each component. There is no page long UI or editor script. Each feature's logical isolation makes it easy to make changes and modify these features.

Basic Setup:

Camera setup or camera rig is composed of two parts.

- Parent Gameobject
- Camera as it child

Parent is responsible for movement(left, right, forward, backward, up and down).

Camera is responsible for the look rotation of the actual camera. Camera stays at 0,0,0 of parent and only changes rotation according to Pan movements from the user.

Components:

PlayerInput.cs:

This component is from Unity's New Input System. It manages the boilerplate setup for the action in CameraControl input actions.

StrategyCameraController.cs:

The brains of the camera controller. It manages the user input, rig's positional limits and camera's pan limit.

Camera's positional limit is represented by two vector3 points. **LowerLeftLimit** and **UpperRightLimit**. These limits are visually represented with a green wire cube marking an area in which the camera can move.

Camera up and down look angle is controlled by **MaxXAngle** and **MinXAngle**. This angle limits how much the camera can look upward and downwards.

FOV limits the zoom of the camera. Its limits are defined by **ZoomInLimit**, **ZoomOutLimit**.

TranslateComponent.cs

This component controls left, right, forward, backward movements of the camera rig.

- Keyboard controls are WASD or Edge hover.
- Touch control: Touch in corners of the screen. Area of active screen corners is EdgeThresholdFactor, which is the percentage of width in pixels on each edge which makes the camera move.

PanComponent.cs

This component controls the look around movement of the camera. Pan up, down, left and right.

- Hold middle mouse button and move mouse to move camera and look around
- Swipe left, right, up, down with two fingers

DragComponent.cs

Look in general direction and click on terrain, move the mouse. World moves with the cursor pulling the point which was clicked.

SelectorComponent.cs

Selection work by clicking on a gameobject with collider and one monobehaviour which implements the interface ISelectable.cs

FollowComponent.cs

If a gameobject is selected, Followcomponent can lock on that component and move such that the selected object stays in the middle of the screen. Camera aligns itself with the object.

TerrainContourFollowComponent.cs

Follow the contour of the terrain and move the rig up and down as the camera moves along the surface. It uses a Raycast directed downward to check for collision with ground underneath and position itself accordingly. You can specify a layermask for terrain to use or you can set a custom layer mask to make the camera move with buildings and other obstacles as well.

ZoomComponent.cs

Zoom changes the FOV of the camera.

- Q and E key are used for zooming on the keyboard.
- Mouse wheel scroll
- Touch pinch gesture control zoom on mobile devices